FINAL REPORT OF INTERNSHIP

ITSimplera Solutions

Enterprise Network Infrastructure, Security & Disaster Recovery Capstone

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Domain:

Network Administration Internship

Tools Used:

GNS3

PuTTY/Solar-PuTTY

EXECUTIVE SUMMARY

This report documents the design, implementation, troubleshooting, testing, and final validation of the Week 08 Enterprise Network Infrastructure, Security & Disaster Recovery Capstone.

The project was implemented entirely in GNS3 and represents a multi-site enterprise environment consisting of four branch locations, a central Headquarters/Core infrastructure, multiple departmental VLANs, a Server Network, a DMZ, and simulated Internet connectivity.

The implemented topology includes Kohat, Lahore, Karachi, and Peshawar branch sites. Each branch is connected to the enterprise core through routed WAN links. The core routing infrastructure uses OSPF as the primary dynamic routing protocol, with multiple OSPF areas used to separate branch networks and the DMZ from the backbone.

The project also includes VLAN segmentation, inter-VLAN connectivity, Layer 2 security features, OSPF authentication and passive interfaces, route summarization, DHCP, NAT, site-to-site IPsec VPN connectivity, Zone-Based Firewall policies, DMZ segmentation, and deployment of an HTTP service using the Networkers Toolkit Toolbox Docker appliance.

A major component of the assignment was troubleshooting. During implementation, multiple issues related to routing, OSPF route advertisement and summarization, DHCP, Docker Toolbox addressing, HTTP service accessibility, VPN configuration interactions, and firewall/service connectivity were investigated and resolved using a systematic troubleshooting methodology.

The final network was validated using Cisco IOS verification commands and end-to-end connectivity testing.

1. INTRODUCTION

The Week 08 assignment is the final capstone assessment of the Network Administration Internship Program. The purpose of the assessment is to combine networking technologies learned during previous internship weeks into a single large-scale enterprise environment.

The assignment requires the intern to design, configure, secure, optimize, troubleshoot, and document an enterprise network entirely in GNS3. The environment represents an organization with multiple branch offices, a central network infrastructure, internal departments, a Data Center/Server environment, a DMZ, Internet connectivity, and security boundaries.

The project was approached as a practical enterprise implementation rather than as isolated configuration tasks. Routing, switching, addressing, security, NAT, VPN connectivity, enterprise services, and troubleshooting were considered as interconnected components.

The final implementation was tested continuously throughout configuration. Where connectivity problems occurred, the troubleshooting process involved identifying the symptom, isolating the affected layer or technology, determining the root cause, applying the required correction, and verifying successful operation.

2. PROJECT OBJECTIVES

The main objectives of the project were to:

- Design a multi-site enterprise network topology in GNS3.
- Implement four branch sites connected to a central routing infrastructure.
- Configure VLAN-based network segmentation.
- Provide inter-VLAN connectivity for departmental networks.
- Configure Layer 2 switching and security mechanisms.
- Implement OSPF as the primary enterprise routing protocol.
- Configure multiple OSPF areas.
- Configure OSPF passive interfaces and authentication where applicable.
- Implement OSPF route summarization.
- Configure redistribution of selected static routes into OSPF.
- Provide redundant routed paths in the enterprise topology.
- Configure DHCP services for client addressing.
- Configure NAT at the Internet edge.
- Implement site-to-site IPsec VPN connectivity.

- Configure Zone-Based Firewall security policies.
- Create and isolate a DMZ network.
- Deploy and test an HTTP service using a Docker Toolbox appliance.
- Verify the complete network using Cisco "show" commands.
- Perform systematic troubleshooting and document the identified problems and their solutions.
- Produce technical documentation containing the topology, addressing plan, VLAN plan, routing design, verification evidence, and troubleshooting report.

3. ENTERPRISE NETWORK ARCHITECTURE

The implemented network consists of the following major components:

3.1 Branch Sites

The enterprise contains four branch locations:

- Kohat
- Lahore
- Karachi
- Peshawar

Each branch contains a local LAN connected to a Layer 2 switch and a branch router. The branch routers connect to the enterprise routed infrastructure.

3.2 Core and Backbone

The central routing infrastructure contains multiple routers connected through point-to-point "/30" networks.

The OSPF backbone is configured as Area 0. Branch areas connect to the backbone through the appropriate routing devices.

3.3 Internal Department Networks

The central enterprise LAN contains separate VLANs for different departments and services.

The implemented departmental networks include:

- IT
- HR

- Management
- Audit
- Sales
- Marketing
- Server Network

3.4 DMZ

A dedicated DMZ network was created using the "192.168.100.0/24" address space.

The DMZ is separated from internal enterprise networks and connected through a dedicated routed security boundary.

3.5 Internet and ISP Simulation

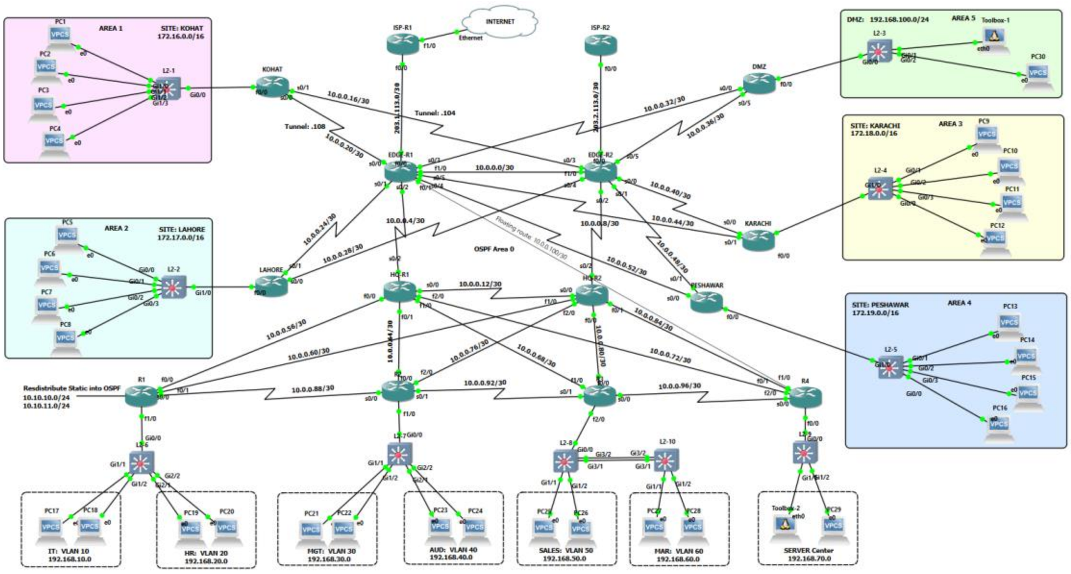
Two ISP routers were used to simulate external connectivity and Internet-facing routing.

Public point-to-point networks were configured between the enterprise edge routers and the simulated ISP environment.

4. FINAL NETWORK TOPOLOGY

The final topology contains the enterprise core, redundant routed links, four branch sites, internal departmental networks, DMZ connectivity, and simulated ISP routers.

The topology uses a large number of "/30" point-to-point networks between routers. This addressing model provides separate Layer 3 networks for individual router links and makes routing relationships easier to identify and troubleshoot.



(Fig1: Network Topology)

5. IP ADDRESSING DESIGN

The addressing plan was divided into three main categories:

1. Branch LAN networks
2. Internal VLAN and departmental networks
3. Point-to-point WAN and routed infrastructure networks

5.1 Branch Site Addressing

Site	OSPF Area	LAN Network	Prefix
Kohat	Area 1	172.16.0.0	/16
Lahore	Area 2	172.17.0.0	/16
Karachi	Area 3	172.18.0.0	/16
Peshawar	Area 4	172.19.0.0	/16

Each branch uses a separate "/16" network for its local LAN.

5.2 DMZ Addressing

Segment	Network	Prefix	OSPF Area
DMZ	192.168.100.0	/24	Area 5

The DMZ is separated from internal enterprise networks and is advertised through OSPF Area 5.

5.3 Departmental and VLAN Addressing

VLAN	Department	Network	Default Gateway
10	IT	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	Management	192.168.30.0/24	192.168.30.1
40	Audit	192.168.40.0/24	192.168.40.1
50	Sales	192.168.50.0/24	192.168.50.1
60	Marketing	192.168.60.0/24	192.168.60.1
70	Data Server	192.168.70.0/24	192.168.70.1

The central enterprise network uses separate "/24" networks for departmental segmentation.

6. COMPLETE ROUTED POINT-TO-POINT ADDRESSING PLAN

The enterprise routed infrastructure uses "/30" subnets from the "10.0.0.0" address space.

Each "/30" network provides two usable host addresses and is suitable for router-to-router point-to-point connections.

No.	Network	Usable Addresses	Broadcast
1	10.0.0.0/30	10.0.0.1 - 10.0.0.2	10.0.0.3
2	10.0.0.4/30	10.0.0.5 - 10.0.0.6	10.0.0.7
3	10.0.0.8/30	10.0.0.9 - 10.0.0.10	10.0.0.11
4	10.0.0.12/30	10.0.0.13 - 10.0.0.14	10.0.0.15
5	10.0.0.16/30	10.0.0.17 - 10.0.0.18	10.0.0.19
6	10.0.0.20/30	10.0.0.21 - 10.0.0.22	10.0.0.23
7	10.0.0.24/30	10.0.0.25 - 10.0.0.26	10.0.0.27
8	10.0.0.28/30	10.0.0.29 - 10.0.0.30	10.0.0.31
9	10.0.0.32/30	10.0.0.33 - 10.0.0.34	10.0.0.35
10	10.0.0.36/30	10.0.0.37 - 10.0.0.38	10.0.0.39
11	10.0.0.40/30	10.0.0.41 - 10.0.0.42	10.0.0.43
12	10.0.0.44/30	10.0.0.45 - 10.0.0.46	10.0.0.47
13	10.0.0.48/30	10.0.0.49 - 10.0.0.50	10.0.0.51
14	10.0.0.52/30	10.0.0.53 - 10.0.0.54	10.0.0.55
15	10.0.0.56/30	10.0.0.57 - 10.0.0.58	10.0.0.59
16	10.0.0.60/30	10.0.0.61 - 10.0.0.62	10.0.0.63
17	10.0.0.64/30	10.0.0.65 - 10.0.0.66	10.0.0.67
18	10.0.0.68/30	10.0.0.69 - 10.0.0.70	10.0.0.71
19	10.0.0.72/30	10.0.0.73 - 10.0.0.74	10.0.0.75
20	10.0.0.76/30	10.0.0.77 - 10.0.0.78	10.0.0.79
21	10.0.0.80/30	10.0.0.81 - 10.0.0.82	10.0.0.83
22	10.0.0.84/30	10.0.0.85 - 10.0.0.86	10.0.0.87
23	10.0.0.88/30	10.0.0.89 - 10.0.0.90	10.0.0.91
24	10.0.0.92/30	10.0.0.93 - 10.0.0.94	10.0.0.95
25	10.0.0.96/30	10.0.0.97 - 10.0.0.97	10.0.0.98
26	10.0.0.100/30	10.0.0.101 - 10.0.0.102	10.0.0.103
27	10.0.0.104/30	10.0.0.105 - 10.0.0.106	10.0.0.107
28	10.0.0.108/30	10.0.0.109 - 10.0.0.110	10.0.0.111

These "/30" networks were allocated to the router-to-router WAN and redundant routed connections shown in the final topology.

6.1 ISP/Public Point-to-Point Addressing

Connection	Network	Prefix	Purpose
EDGE-R1 to ISP-R1	203.1.113.0	/30	Internet/ISP connectivity
EDGE-R2 to ISP-R2	203.2.113.0	/30	Internet/ISP connectivity

The public addressing was used to simulate enterprise edge connectivity to two separate ISP routers.

7. LAYER 2 IMPLEMENTATION AND SECURITY

The Layer 2 infrastructure was configured to provide access connectivity and VLAN segmentation.

The following features were implemented or configured where supported by the selected GNS3 devices:

- Access port configuration
- VLAN assignment
- Trunk links
- Spanning Tree Protocol protection
- PortFast on end-device-facing ports
- BPDU Guard
- Root Guard where required
- Port Security
- Sticky MAC addressing
- DHCP Snooping configuration

These features were intended to improve access-layer security and prevent common Layer 2 problems such as unauthorized device access, rogue DHCP responses, accidental topology changes, and STP-related failures.

```
Switch#show spann sum
Switch is in rapid-pvst mode
Root bridge for: VLAN0001, VLAN0010, VLAN0020
Extended system ID                      is enabled
Portfast Default                        is disabled
Portfast Edge BPDU Guard Default        is disabled
Portfast Edge BPDU Filter Default       is disabled
Loopguard Default                       is disabled
PVST Simulation Default                 is enabled but inactive in rapid-pvst mode
Bridge Assurance                        is enabled
EtherChannel misconfig guard            is enabled
Configured Pathcost method used is short
UplinkFast                              is disabled
BackboneFast                            is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                     0         0        0         11         11
VLAN0010                     0         0        0          3          3
VLAN0020                     0         0        0          3          3
---------------------- -------- --------- -------- ---------- ----------
3 vlans                      0         0        0         17         17
```

(Figure 2: Spanning Tree Verification)

```
Switch#show port-s
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
                (Count)       (Count)          (Count)
---------------------------------------------------------------------------
      Gi1/1              2            0                  0         Restrict
      Gi1/2              2            0                  0         Restrict
      Gi2/1              2            0                  0         Restrict
      Gi2/2              2            0                  0         Restrict
---------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096
Switch#
```

(Figure 3: Port Security Verification)

```
Switch#show ip dhcp snoo
Switch DHCP snooping is enabled
Switch DHCP gleaning is disabled
DHCP snooping is configured on following VLANs:
10,20
DHCP snooping is operational on following VLANs:
10,20
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is enabled
   circuit-id default format: vlan-mod-port
   remote-id: 0c24.cb60.0000 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                  Trusted    Allow option    Rate limit (pps)
-----------------------    -------    ------------    ----------------
GigabitEthernet0/0         yes        yes             unlimited
```

Figure 4: DHCP Snooping Verification

8. ROUTING DESIGN

OSPF was implemented as the primary enterprise routing protocol.

The network was divided into multiple OSPF areas to logically separate branch networks from the backbone.

8.1 OSPF Area Design

OSPF AREA	LOCATION
Area 0	Enterprise Backbone / Headquarter
Area 1	Kohat
Area 2	Lahore
Area 3	Karachi
Area 4	Peshawar
Area 5	DMZ

Area 0 functions as the OSPF backbone. Branch and DMZ networks are connected to the backbone through their respective routed connections.

8.2 OSPF Features

The following OSPF features were configured and tested:

- Multi-area OSPF
- OSPF neighbor formation
- Passive interfaces
- OSPF route advertisement
- OSPF authentication where configured
- Route summarization
- ABR connectivity between backbone and non-backbone areas
- Redistribution of selected static routing information

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
203.1.113.2       1   FULL/DR         00:00:30    203.1.113.2     FastEthernet0/0
3.3.3.3           0   FULL/  -        00:00:39    10.0.0.6        Serial0/2
2.2.2.2           1   FULL/DR         00:00:36    10.0.0.2        FastEthernet1/0
172.16.0.1        0   FULL/  -        00:00:32    10.0.0.22       Serial0/0
172.17.0.1        0   FULL/  -        00:00:32    10.0.0.26       Serial0/1
172.18.0.1        0   FULL/  -        00:00:30    10.0.0.46       Serial0/5
172.19.0.1        0   FULL/  -        00:00:31    10.0.0.54       Serial0/4
192.168.100.1     0   FULL/  -        00:00:38    10.0.0.34       Serial0/3
EDGE-R1#
```

(Figure 5: OSPF Neighbor Relationships)

```
EDGE-R1#show ip ospf int br
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
NV0          1     0               0.0.0.0/0          1     P2P   0/0
Fa0/0        1     0               203.1.113.1/30     10    BDR   1/1
Se0/2        1     0               10.0.0.5/30        64    P2P   1/1
Fa1/0        1     0               10.0.0.1/30        1     BDR   1/1
Se0/0        1     1               10.0.0.21/30       64    P2P   1/1
Se0/1        1     2               10.0.0.25/30       64    P2P   1/1
Se0/5        1     3               10.0.0.45/30       64    P2P   1/1
Se0/4        1     4               10.0.0.53/30       64    P2P   1/1
Se0/3        1     5               10.0.0.33/30       64    P2P   1/1
EDGE-R1#
```

(Figure 6: OSPF Interface)

```
     203.2.113.0/30 is subnetted, 1 subnets
O       203.2.113.0 [110/11] via 10.0.0.2, 02:14:05, FastEthernet1/0
O    192.168.60.0/24 [110/66] via 10.0.0.6, 02:14:35, Serial0/2
O    192.168.10.0/24 [110/75] via 10.0.0.6, 02:14:35, Serial0/2
O    172.17.0.0/16 [110/65] via 10.0.0.26, 02:14:35, Serial0/1
O    172.16.0.0/16 [110/65] via 10.0.0.22, 02:14:35, Serial0/0
O    172.19.0.0/16 [110/65] via 10.0.0.54, 02:14:35, Serial0/4
O    172.18.0.0/16 [110/65] via 10.0.0.46, 02:14:35, Serial0/5
O    192.168.20.0/24 [110/75] via 10.0.0.6, 02:14:35, Serial0/2
     10.0.0.0/8 is variably subnetted, 33 subnets, 2 masks
O       10.0.0.8/30 [110/65] via 10.0.0.2, 02:14:05, FastEthernet1/0
O       10.0.0.12/30 [110/128] via 10.0.0.6, 02:14:35, Serial0/2
O       10.0.0.24/29 is a summary, 02:14:45, Null0
O       10.0.0.28/30 [110/128] via 10.0.0.26, 02:14:35, Serial0/1
O       10.0.0.16/30 [110/128] via 10.0.0.22, 02:14:35, Serial0/0
O       10.0.0.16/29 is a summary, 02:14:45, Null0
O       10.0.0.40/30 [110/128] via 10.0.0.46, 02:14:37, Serial0/5
O       10.0.0.40/29 is a summary, 02:14:47, Null0
O       10.0.0.32/29 is a summary, 02:14:47, Null0
O       10.0.0.36/30 [110/128] via 10.0.0.34, 02:14:37, Serial0/3
O       10.0.0.56/30 [110/74] via 10.0.0.6, 02:14:37, Serial0/2
O       10.0.0.60/30 [110/76] via 10.0.0.6, 02:14:37, Serial0/2
O       10.0.0.48/30 [110/128] via 10.0.0.54, 02:14:37, Serial0/4
O       10.0.0.48/29 is a summary, 02:14:49, Null0
O       10.0.0.72/30 [110/65] via 10.0.0.6, 02:14:39, Serial0/2
O       10.0.0.76/30 [110/75] via 10.0.0.6, 02:14:39, Serial0/2
O       10.0.0.64/30 [110/74] via 10.0.0.6, 02:14:40, Serial0/2
O       10.0.0.68/30 [110/65] via 10.0.0.6, 02:14:40, Serial0/2
O       10.0.0.88/30 [110/138] via 10.0.0.6, 02:14:10, Serial0/2
O       10.0.0.92/30 [110/129] via 10.0.0.6, 02:14:41, Serial0/2
O       10.0.0.80/30 [110/75] via 10.0.0.6, 02:14:41, Serial0/2
O       10.0.0.84/30 [110/75] via 10.0.0.6, 02:14:42, Serial0/2
O       10.0.0.104/30 [110/11239] via 10.0.0.22, 02:14:43, Serial0/0
O       10.0.0.96/30 [110/129] via 10.0.0.6, 02:14:13, Serial0/2
O    192.168.50.0/24 [110/66] via 10.0.0.6, 02:14:44, Serial0/2
O    192.168.70.0/24 [110/66] via 10.0.0.6, 02:14:45, Serial0/2
O    192.168.100.0/24 [110/65] via 10.0.0.34, 02:14:46, Serial0/3
```

(Figure 7: OSPF Routing Table)

9. OSPF ROUTE SUMMARIZATION

Route summarization was used to reduce the number of individual networks advertised between OSPF areas where appropriate.

The enterprise topology contains a significant number of "/30" point-to-point networks. Instead of advertising every individual network across all areas, selected contiguous networks were summarized at the appropriate area boundary.

The purpose of summarization was to:

- Reduce routing table size.
- Reduce unnecessary routing information between areas.
- Simplify routing information.
- Improve scalability.

During implementation, route summarization required verification because previously learned specific routes remained visible until routing information was updated and recalculated. The routing table and OSPF advertisements were checked to distinguish between local intra-area routes and summarized inter-area advertisements.

```
EDGE-R1#show ip route | include /29
O       10.0.0.24/29 is a summary, 02:22:21, Null0
O       10.0.0.16/29 is a summary, 02:22:21, Null0
O       10.0.0.40/29 is a summary, 02:22:21, Null0
O       10.0.0.32/29 is a summary, 02:22:21, Null0
O       10.0.0.48/29 is a summary, 02:22:21, Null0
```

(Figure 8: Route Summarization Evidence)

10. STATIC ROUTE REDISTRIBUTION

Static routing information was redistributed into OSPF where required so that the relevant networks became reachable throughout the OSPF domain.

The redistribution configuration was verified using "show ip route & show ip protocols.

```
EDGE-R1#show ip rout ospf | include /24
O    192.168.60.0/24 [110/66] via 10.0.0.6, 02:36:21, Serial0/2
O    192.168.10.0/24 [110/75] via 10.0.0.6, 02:36:21, Serial0/2
O    192.168.20.0/24 [110/75] via 10.0.0.6, 02:36:21, Serial0/2
O E2    10.10.10.0/24 [110/20] via 10.0.0.6, 00:04:34, Serial0/2
O E2    10.10.11.0/24 [110/20] via 10.0.0.6, 00:04:34, Serial0/2
O    192.168.50.0/24 [110/66] via 10.0.0.6, 02:36:21, Serial0/2
O    192.168.70.0/24 [110/66] via 10.0.0.6, 02:36:21, Serial0/2
O    192.168.100.0/24 [110/65] via 10.0.0.34, 02:36:21, Serial0/3
```

(Figure 9: Route Redistribution Verification)

11. REDUNDANCY AND ROUTED PATH DESIGN

The routed topology was designed with multiple inter-router connections.

Redundant links were included between core and edge routing devices so that the routing protocol could select available paths according to the topology and routing metrics.

OSPF was used as the primary dynamic routing mechanism for route selection and convergence.

The redundancy design was verified by examining routing paths and router connectivity.

```
                Router Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         1758        0x80000006 0x00F5A8 2
2.2.2.2         2.2.2.2         1767        0x80000006 0x002D5F 3
172.16.0.1      172.16.0.1      1819        0x80000007 0x00E655 5

                Summary Net Link States (Area 1)

Link ID         ADV Router      Age         Seq#       Checksum
10.0.0.0        1.1.1.1         1765        0x80000005 0x00D854
10.0.0.0        2.2.2.2         1768        0x80000005 0x00BA6E
10.0.0.4        1.1.1.1         1765        0x80000005 0x0029C0
10.0.0.4        2.2.2.2         1768        0x80000005 0x0015CF
10.0.0.8        1.1.1.1         1765        0x80000005 0x000BD9
10.0.0.8        2.2.2.2         1768        0x80000005 0x00E2FE
10.0.0.12       1.1.1.1         1765        0x80000005 0x005B46
10.0.0.12       2.2.2.2         1768        0x80000005 0x004755
10.0.0.24       1.1.1.1         1766        0x80000005 0x004891
10.0.0.24       2.2.2.2         1770        0x80000005 0x002AAB
10.0.0.32       1.1.1.1         1769        0x80000005 0x00F7D9
10.0.0.32       2.2.2.2         1772        0x80000005 0x00D9F3
10.0.0.40       1.1.1.1         1770        0x80000005 0x00A722
10.0.0.40       2.2.2.2         1773        0x80000005 0x00893C
10.0.0.48       1.1.1.1         1771        0x80000005 0x00576A
10.0.0.48       2.2.2.2         1774        0x80000005 0x003984
10.0.0.56       1.1.1.1         1772        0x80000005 0x008328
10.0.0.56       2.2.2.2         1774        0x80000005 0x006F37
10.0.0.60       1.1.1.1         1772        0x80000005 0x006F36
10.0.0.60       2.2.2.2         1775        0x80000005 0x005B45
10.0.0.64       1.1.1.1         1773        0x80000005 0x003370
10.0.0.64       2.2.2.2         1776        0x80000005 0x001F7F
10.0.0.68       1.1.1.1         1774        0x80000005 0x00B0F7
10.0.0.68       2.2.2.2         1777        0x80000005 0x009C07
10.0.0.72       1.1.1.1         1776        0x80000005 0x00881C
10.0.0.72       2.2.2.2         1779        0x80000005 0x00742B
 --More--
```

(Figure 10: Redundant Routed Links)

12. DHCP IMPLEMENTATION

DHCP was configured to provide automatic IPv4 addressing to client devices.

The DHCP implementation was verified by checking:

- DHCP pool information.
- DHCP bindings.
- Client IP address assignment.
- Client connectivity after receiving an address.

During testing, a DHCP-related issue was encountered and resolved. After correction, the client successfully obtained an address and network connectivity was restored.

```
Pool IT-LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.10.1         192.168.10.1     - 192.168.10.254    0

Pool HR-LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.20.1         192.168.20.1     - 192.168.20.254    0
```

(Figure 11: DHCP Pools)

13. NAT IMPLEMENTATION

NAT was configured on the enterprise Internet edge to allow appropriate translation between private enterprise addressing and the simulated external network.

Internal and external NAT interfaces were configured according to their location in the topology.

NAT operation was verified using:

- "show ip nat translations"
- "show ip nat statistics"

The NAT configuration was also considered when configuring VPN traffic because VPN traffic must not be incorrectly translated when it matches protected IPsec traffic.

```
EDGE-R1#show ip nat trans
Pro Inside global      Inside local       Outside local      Outside global
icmp 203.1.113.1:1     172.17.0.1:0       203.1.113.2:0      203.1.113.2:1
icmp 203.1.113.1:2     192.168.10.1:0     203.1.113.2:0      203.1.113.2:2
EDGE-R1#
```

(Figure 12: NAT Translation Table)

```
EDGE-R1#show ip nat sta
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Outside interfaces:
  FastEthernet0/0
Inside interfaces:
  Serial0/0, FastEthernet0/1, Serial0/1, Serial0/2, Serial0/3, Serial0/4
  Serial0/5, FastEthernet1/0
Hits: 30  Misses: 0
CEF Translated packets: 30, CEF Punted packets: 0
Expired translations: 3
Dynamic mappings:
-- Inside Source
[Id: 1] access-list NAT-ACL interface FastEthernet0/0 refcount 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
```

(Figure 13: NAT Statistics)

14. SITE-TO-SITE IPSEC VPN

Site-to-site IPsec VPN connectivity was configured between the required enterprise routers.

The configuration included the following components:

- IKE/ISAKMP Phase 1 policy.
- AES encryption.
- Pre-shared key authentication.
- Diffie-Hellman group configuration.
- Phase 2 transform set.
- Crypto ACL defining interesting traffic.
- Crypto map configuration.
- Application of the crypto map to the appropriate WAN interface.

During VPN implementation, existing and newly created VPN configurations had to be carefully separated to avoid conflicts between peer-specific crypto configurations.

VPN verification was performed using:

- "show crypto isakmp sa"

- "show crypto ipsec sa"

```
KOHAT#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.0.21       10.0.0.22       QM_IDLE           1002    0 ACTIVE
10.0.0.18       10.0.0.17       QM_IDLE           1001    0 ACTIVE
```

(Figure 14: IKE/ISAKMP Verification)

```
KOHAT#show crypto ipsec sa

interface: Serial0/0
    Crypto map tag: VPN-EDGE1, local addr 10.0.0.22

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.0.0.22/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (10.0.0.21/255.255.255.255/47/0)
   current_peer 10.0.0.21 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4, #pkts encrypt: 4, #pkts digest: 4
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

     local crypto endpt.: 10.0.0.22, remote crypto endpt.: 10.0.0.21
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0
     current outbound spi: 0x5E0CFD7D(1577909629)
```

(Figure 15: IPsec Verification)

15. ZONE-BASED FIREWALL

Zone-Based Firewall policies were configured to control traffic between security zones.

The firewall design uses logical security zones representing internal, external, and protected network boundaries.

The configuration includes:

- Security zones.
- Zone pairs.
- Class maps.
- Policy maps.
- Traffic inspection policies.

The firewall was tested to ensure that permitted traffic could pass while restricted traffic remained blocked according to the configured policy.

```
EDGE-R1#show zone sec
zone self
  Description: System defined zone


zone INSIDE
  Member Interfaces:
    Tunnel0
    Serial0/0
    FastEthernet0/1
    Serial0/1
    Serial0/2
    Serial0/4
    Serial0/5
    FastEthernet1/0


zone OUTSIDE
  Member Interfaces:
    FastEthernet0/0


zone DMZ
  Member Interfaces:
    Serial0/3
```

(Figure 16: Zone Security)

```
EDGE-R1#show zone-pa sec
Zone-pair name ZP-INSIDE-OUT
    Source-Zone INSIDE  Destination-Zone OUTSIDE
    service-policy PM-INSIDE-OUT
Zone-pair name ZP-INSIDE-DMZ
    Source-Zone INSIDE  Destination-Zone DMZ
    service-policy PM-INSIDE-DMZ
Zone-pair name ZP-OUTSIDE-DMZ
    Source-Zone OUTSIDE  Destination-Zone DMZ
    service-policy PM-OUTSIDE-DMZ
```

(Figure 17: Zone-Pair Verification)

```
EDGE-R1#show policy-map type inspect
  Policy Map type inspect PM-OUTSIDE-DMZ
    Class CM-OUTSIDE-DMZ
      Inspect
    Class class-default

  Policy Map type inspect PM-INSIDE-OUT
    Class CM-INSIDE-OUT
      Inspect
    Class class-default

  Policy Map type inspect PM-INSIDE-DMZ
    Class CM-INSIDE-DMZ
      Inspect
    Class class-default
```

(Figure 18: Policy Map Verification)

16. DMZ IMPLEMENTATION

The DMZ is connected to the routed enterprise environment and separated from internal networks through the configured security architecture.

The purpose of the DMZ is to provide a separate network zone for services requiring controlled accessibility without placing those services directly inside the internal enterprise LAN.

DMZ connectivity and routing were verified using interface, routing, and end-to-end connectivity testing.

17. TOOLBOX SERVER AND HTTP SERVICE

The Networkers Toolkit Docker appliance was deployed in the Server Network.

The Toolbox was manually assigned a valid IP address from the Server Network.

Initially, service troubleshooting focused on the difference between a normal virtual machine environment and the Docker-based Toolbox environment. Commands such as "systemctl" and the expected service-management commands did not behave as initially assumed.

After identifying the correct environment behavior and completing the network addressing configuration, the HTTP service became reachable from the client.

The final HTTP connectivity test confirmed that the client could successfully access the Toolbox-hosted web service.

```
root@Toolbox-1:~# ip addr add 192.168.100.10/24 dev eth0
root@Toolbox-1:~# ip link set eth0 up
root@Toolbox-1:~# ip route add default via 192.168.100.1
root@Toolbox-1:~#
root@Toolbox-1:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
6: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 02:42:16:a5:72:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::42:16ff:fea5:7200/64 scope link
       valid_lft forever preferred_lft forever
root@Toolbox-1:~#
root@Toolbox-1:~# ip route
default via 192.168.100.1 dev eth0
192.168.100.0/24 dev eth0 proto kernel scope link src 192.168.100.10
root@Toolbox-1:~#
```

(Figure 19: Toolbox IP Configuration)

```
curl: (7) Failed to connect to 192.168.100.2 port 8080: Connection refused
root@Toolbox-1:~#
root@Toolbox-1:~# curl http://192.168.100.2
<!DOCTYPE html>
<html>
<head>
<title>Networkers' Toolkit</title>
</head>
<body style="background-color: #003c4f; color: white;">
<table border="0" cellspacing="0" cellpadding="0" align="center">
<tbody>
<tr>
<td style="text-align: center;"><img src="gns3.png" /></td>
</tr>
<tr>
<td>
<h1 style="text-align: center;">Networkers' Toolkit</h1>
</td>
</tr>
<tr>
<td>
<strong>WWW:</strong> Files located at /var/www/html<br /> <strong>FTP:</strong> You get t
o /root after logging in but you're not chrooted<br /><strong>TFTP:</strong> You can place
 your files in /tftpboot; upload is permitted<br /><strong>DHCP:</strong> The DHCP server
is installed but not configured to avoid any possible conflict<br /><strong>Syslog server
&amp; SNMP trap receiver:</strong> They both log to /var/log/syslog
</td>
</tr>
</tbody>
```

(Figure 20: HTTP Service Verification)

18. SECURE DEVICE MANAGEMENT

Secure device-management features were configured where supported and required in the implementation.

The security configuration included relevant controls such as:

- SSH access.
- Local authentication.
- AAA where configured.
- Administrative privilege configuration.
- Secure VTY access.
- Password protection.
- Password encryption.
- Login security controls.
- Session timeout.
- MOTD/banner configuration.
- Restriction of insecure management methods where configured.

```
LAHORE#ssh -l admin 10.0.0.1

Password:

*** AUTHORIZED ACCESS ONLY ***


EDGE-R1#
EDGE-R1#show ssh
Connection Version Mode Encryption  Hmac         State                 Username
98         1.99     IN   aes128-cbc  hmac-sha1    Session started       admin
98         1.99     OUT  aes128-cbc  hmac-sha1    Session started       admin
%No SSHv1 server connections running.
EDGE-R1#
EDGE-R1#show user
    Line       User       Host(s)              Idle       Location
   0 con 0     admin      idle                 00:00:41
* 98 vty 0     admin      idle                 00:00:00 10.0.0.26
```

(Figure 21: SSH and User Verification)

19. TROUBLESHOOTING METHODOLOGY

Troubleshooting was performed systematically rather than by changing configurations randomly.

The following process was used:

Step 1 – Identify the Symptom

The affected device or service was identified. Examples included missing DHCP addressing, unavailable HTTP access, unexpected routing information, or a VPN configuration problem.

Step 2 – Isolate the Technology Layer

The issue was classified according to the affected technology:

- Layer 1/Interface
- Layer 2/VLAN
- Layer 3/Routing
- OSPF
- DHCP
- NAT
- VPN
- Firewall
- Enterprise Service

Step 3 – Verify Current Operation

Relevant "show" commands, routing tables, interface status, and connectivity tests were used to determine the current state.

Step 4 – Identify Root Cause

The configuration was compared with the intended topology and addressing design to identify the actual source of the problem.

Step 5 – Apply Correction

Only the configuration responsible for the issue was corrected.

Step 6 – Verify the Fix

The service or connectivity test was repeated after the correction. Before-and-after evidence was captured where possible.

20. TROUBLESHOOTING REPORT

ISSUE SYMPTOMS	ROOT CAUSE	FIX	VERIFICATION
OSPF neighbors were not forming.	OSPF configuration or network/area parameters were mismatched.	Corrected OSPF network statements and area assignments.	"show ip ospf neighbor" confirmed FULL adjacencies
OSPF routes were missing from the routing table.	Networks were not being properly advertised by OSPF.	Corrected OSPF advertisements and verified participating interfaces.	"show ip route ospf" displayed the expected routes
OSPF hello packets were being sent but no neighbor appeared.	The connected interfaces had an OSPF parameter/configuration mismatch.	Matched OSPF area and interface configuration on both ends.	OSPF neighbors successfully reached FULL state
Area-based OSPF routes were not appearing as expected.	Area assignments and ABR advertisements required correction	Corrected area configuration and verified ABR connectivity	"show ip ospf database" and "show ip route ospf" confirmed area routes
OSPF summarized route was present together with specific /30 routes	Specific routes were still being learned through existing OSPF LSAs	Verified summarization at the correct ABR and allowed OSPF to reconverge	Routing table showed the intended summary route after reconvergence
VLAN 10 SVI was down/broken	STP Root Guard was incorrectly applied and placed the VLAN in an inconsistent state	Removed the incorrect Root Guard configuration and corrected STP settings	"show ip interface brief" showed VLAN 10 as up/up.
Inter-VLAN communication was not working	VLAN/SVI configuration or Layer-2 connectivity was incorrect	Corrected VLAN membership, trunking, and SVI configuration	Hosts in different VLANs successfully communicated
Trunk link was not carrying required VLANs	Trunk configuration/ native or allowed VLAN configuration was incorrect	Corrected trunk configuration and allowed VLANs	"show interfaces trunk" confirmed the required VLANs were active
STP caused an unexpected port state	STP protection configuration was applied to an inappropriate interface	Reviewed and corrected Root Guard/PortFast/BPDU Guard placement	"show spanning-tree" showed the expected STP state
DHCP clients were not receiving addresses	DHCP forwarding/relay configuration was missing or incorrect	Configured the required "ip helper-address" and corrected DHCP settings	VPCs successfully received IP addresses
Toolbox could not be reached from the VPC	Toolbox IP addressing was not correctly configured.	Assigned the correct IP address, subnet mask, and gateway	VPC successfully pinged the Toolbox IP.
HTTP website was not accessible from the VPC.	Web service on the Toolbox was not running/configured correctly	Configured the Toolbox web service and verified its listening interface/port	Browser successfully opened the HTTP webpage from the VPC.
IPsec VPN was not establishing	ISAKMP/IKE peer, authentication, encryption, or pre-shared-key parameters were mismatched	Matched Phase 1 parameters and corrected the peer configuration	"show crypto isakmp sa" showed an established IKE state

NAT translations were not appearing	NAT interfaces or NAT ACL/rules were incorrectly configured	Corrected inside/outside interfaces and NAT matching rules	"show ip nat translations" displayed active translations
Internet-bound traffic was not reaching the ISP	Default route or upstream next-hop configuration was missing/incorrect	Corrected the default route toward the ISP	"show ip route" displayed the correct default route and end-to-end ping succeeded.

21. FINAL CONNECTIVITY TESTING

The following functional tests were performed as part of final validation:

- End-device to default gateway ping.
- Inter-VLAN connectivity testing.
- Branch-to-core connectivity.
- Branch-to-branch connectivity through OSPF.
- OSPF neighbor verification.
- OSPF route verification.
- DHCP address assignment.
- NAT translation verification.
- Site-to-site VPN verification.
- DMZ reachability testing.
- Zone-Based Firewall policy testing.
- Toolbox reachability.
- HTTP service access.

22. FINAL RESULTS

The final Week 08 topology successfully integrated multiple enterprise networking technologies into one GNS3 environment.

The completed implementation includes four branch networks, departmental VLAN segmentation, a Server Network, a DMZ, simulated ISP connectivity, multi-area OSPF routing, route summarization, DHCP, NAT, IPsec VPN connectivity, Zone-Based Firewall policies, and HTTP service deployment.

The project also required significant troubleshooting during implementation. Problems were approached systematically by identifying symptoms, isolating the affected technology, determining the root cause, correcting the configuration, and verifying successful operation.

Successful DHCP operation and HTTP access to the Toolbox server confirmed that enterprise network services were functioning correctly. Routing and OSPF verification confirmed connectivity between the major network segments. NAT, VPN, firewall, and DMZ components were verified using their respective operational commands and connectivity tests.

23. CONCLUSION

The Week 08 Enterprise Network Infrastructure, Security & Disaster Recovery Capstone provided practical experience in the implementation and troubleshooting of a large multi-site enterprise network.

The project required the integration of routing, switching, VLAN segmentation, OSPF, NAT, VPN, firewall policies, DHCP, DMZ connectivity, and enterprise services into a single environment.

One of the most important aspects of the project was troubleshooting. Rather than only configuring devices, problems had to be identified from symptoms and resolved through systematic analysis. Routing tables, OSPF neighbors, interface status, DHCP information, VPN security associations, firewall policies, and end-to-end connectivity tests were used to verify network operation.

The final implementation demonstrates practical understanding of enterprise network design, IP addressing, routing, security, troubleshooting, and technical documentation.